

Instalación del servidor Tomcat

Despliegue de Aplicaciones Web

En la presente práctica tenemos como objetivo realizar y documentar lo necesario para la puesta en marcha de un servidor de aplicaciones usando Tomcat sobre Ubuntu Server. Necesitamos para ello realizar la instalación y configuración del software necesario:

1. Instala el servidor Tomcat 11

```
ubuntu@ip-172-31-40-208: ~$ sudo apt install openjdk-21-jdk -y
```

```
ubuntu@ip-172-31-40-208: /tmp$ wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.22/bin/apache-tomcat-11.0.22.tar.gz
--2026-05-08 11:26:14-- https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.22/bin/apache-tomcat-11.0.22.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14370214 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-11.0.22.tar.gz'

apache-tomcat-11.0.22.tar.gz 100%[=====] 13.70M --.-KB/s in 0.03s

2026-05-08 11:26:15 (423 MB/s) - 'apache-tomcat-11.0.22.tar.gz' saved [14370214/14370214]

ubuntu@ip-172-31-40-208: /tmp$
```

```
ubuntu@ip-172-31-40-208: /tmp$ sudo tar -xvf apache-tomcat-11.0.22.tar.gz -C /opt/
ubuntu@ip-172-31-40-208: /tmp$ sudo mv /opt/apache-tomcat-11.0.22 /opt/tomcat
ubuntu@ip-172-31-40-208: /tmp$
```

2. Configura y fija para el sistema las variables de entorno para la correcta ejecución de Tomcat.

```
GNU nano 8.7.1 /etc/environment *
PATH="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
CATALINA_HOME="/opt/tomcat"
JAVA_HOME="/usr/lib/jvm/java-21-openjdk-amd64"
```

```
ubuntu@ip-172-31-40-208: /tmp$ sudo nano /etc/environment
ubuntu@ip-172-31-40-208: /tmp$ source /etc/environment
```

```
ubuntu@ip-172-31-40-208: /opt/tomcat$ ls
BUILDING.txt CONTRIBUTING.md LICENSE NOTICE README.md RELEASE-NOTES RUNNING.txt bin conf lib logs temp webapps work
ubuntu@ip-172-31-40-208: /opt/tomcat$ sudo chmod +x /bin/*.sh
```

```
ubuntu@ip-172-31-40-208:/opt/tomcat$ sudo /opt/tomcat/bin/startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr/lib/jvm/java-21-openjdk-amd64
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Tomcat started.
ubuntu@ip-172-31-40-208:/opt/tomcat$
```

3. Gestiona el servicio instalado:

- Comprueba el estado, para, inicia.
- Deshabilita el arranque automático, reinicia la máquina y comprueba el estado.
- Habilita, inicia y comprueba el estado.

```
ubuntu@ip-172-31-40-208:/opt/tomcat$ sudo nano /etc/systemd/system/tomcat.service
ubuntu@ip-172-31-40-208:/opt/tomcat$
```

```
GNU nano 8.7.1 /etc/systemd/system/tomcat.service *
[Unit]
Description=Apache Tomcat 11
After=network.target

[Service]
Type=forking

# Variables de entorno para el servicio
Environment=JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
Environment=CATALINA_PID=/opt/tomcat/temp/tomcat.pid
Environment=CATALINA_HOME=/opt/tomcat
Environment=CATALINA_BASE=/opt/tomcat

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh

User=root
Group=root
RestartSec=10
Restart=always

[Install]
WantedBy=multi-user.target
```

```
ubuntu@ip-172-31-40-208:/opt/tomcat$ sudo systemctl daemon-reload
ubuntu@ip-172-31-40-208:/opt/tomcat$
```

```
ubuntu@ip-172-31-40-208:/opt/tomcat$ sudo systemctl daemon-reload
ubuntu@ip-172-31-40-208:/opt/tomcat$ sudo systemctl start tomcat
ubuntu@ip-172-31-40-208:/opt/tomcat$ sudo systemctl status tomcat
● tomcat.service - Apache Tomcat 11
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-05-08 11:43:19 UTC; 2min 35s ago
  Invocation: fb3063a762d545d3bed1a4da849b2c6d
    Main PID: 4592 (java)
      Tasks: 29 (limit: 627)
     Memory: 77.2M (peak: 79.4M)
        CPU: 4.809s
    CGroup: /system.slice/tomcat.service
            └─4592 /usr/lib/jvm/java-21-openjdk-amd64/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava

May 08 11:43:09 ip-172-31-40-208 systemd[1]: tomcat.service: Control process exited, code=exited, status=1/FAILURE
May 08 11:43:09 ip-172-31-40-208 systemd[1]: tomcat.service: Failed with result 'exit-code'.
May 08 11:43:09 ip-172-31-40-208 systemd[1]: tomcat.service: Consumed 4.422s CPU time over 12.432s wall clock time, 71M memory peak.
May 08 11:43:19 ip-172-31-40-208 systemd[1]: tomcat.service: Scheduled restart job, restart counter is at 6.
May 08 11:43:19 ip-172-31-40-208 systemd[1]: Starting tomcat.service - Apache Tomcat 11...
May 08 11:43:19 ip-172-31-40-208 startup.sh[4582]: Existing PID file found during start.
May 08 11:43:19 ip-172-31-40-208 startup.sh[4582]: Removing/clearing stale PID file.
May 08 11:43:19 ip-172-31-40-208 startup.sh[4582]: Tomcat started.
May 08 11:43:19 ip-172-31-40-208 systemd[1]: Started tomcat.service - Apache Tomcat 11.
```

4. Investiga sobre la estructura de gestor directorios de Tomcat en las dos opciones de instalación indicadas y explica brevemente que contiene cada directorio de Tomcat.
5. En el terminal, averigua los puertos TCP y/o UDP que están a la escucha. Cambia el puerto en el que escucha Tomcat, reinicia el servicio y vuelve a comprobar los puertos.
6. Mediante el navegador accede al servidor. Comprobar los logs del mismo.

Browser address bar: No es seguro 34.238.143.100:8080

Navigation: Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/11.0.22

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:
[Security Considerations How-To](#)
[Manager Application How-To](#)
[Clustering/Session Replication How-To](#)

[Server Status](#)
[Manager App](#)
[Host Manager](#)

Developer Quick Start

[Tomcat Setup](#)
[First Web Application](#)

[Realms & AAA](#)
[JDBC DataSources](#)

[Examples](#)

[Servlet Specifications](#)
[Tomcat Versions](#)

Managing Tomcat

For security, access to the **manager webapp** is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 11.0 access to the manager application is split between different users.
[Read more...](#)

Release Notes
Changelog
Migration Guide
Security Notices

Documentation

[Tomcat 11.0 Documentation](#)
[Tomcat 11.0 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

- [Tomcat 11.0 Bug Database](#)
- [Tomcat 11.0 JavaDocs](#)
- [Tomcat 11.0 Git Repository at GitHub](#)

Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [taglibs-user](#)
User support and discussion for [Apache Taglibs](#)
- [tomcat-dev](#)
Development mailing list, including commit messages

Other Downloads Tomcat Connectors Tomcat Native Taglibs Deployer	Other Documentation Tomcat Connectors mod_ik Documentation Tomcat Native Deployer	Get Involved Overview Source Repositories Mailing Lists Wiki	Miscellaneous Contact Legal Sponsorship Thanks	Apache Software Foundation Who We Are Heritage Apache Home Resources
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```
ubuntu@ip-172-31-40-208: /opt/tomcat
GNU nano 8.7.1 /opt/tomcat/conf/tomcat-users.xml *
to operate the "/manager/html" web application. If you wish to use this app,
you must define such a user - the username and password are arbitrary.

Built-in Tomcat manager roles:
- manager-gui - allows access to the HTML GUI and the status pages
- manager-script - allows access to the HTTP API and the status pages
- manager-jmx - allows access to the JMX proxy and the status pages
- manager-status - allows access to the status pages only

The users below are wrapped in a comment and are therefore ignored. If you
wish to configure one or more of these users for use with the manager web
application, do not forget to remove the <!-- ... --> that surrounds them. You
will also need to set the passwords to something appropriate.
-->
<!--
<user username="admin" password="<must-be-changed>" roles="manager-gui"/>
<user username="robot" password="<must-be-changed>" roles="manager-script"/>
-->
<!--
The sample user and role entries below are intended for use with the
examples web application. They are wrapped in a comment and thus are ignored
when reading this file. If you wish to configure these users for use with the
examples web application, do not forget to remove the <!-- ... --> that surrounds
them. You will also need to set the passwords to something appropriate.
-->
<!--
<role rolename="tomcat"/>
<role rolename="role1"/>
<user username="tomcat" password="<must-be-changed>" roles="tomcat"/>
<user username="both" password="<must-be-changed>" roles="tomcat,role1"/>
<user username="role1" password="<must-be-changed>" roles="role1"/>
-->
<role rolename="manager-gui"/>
<role rolename="admin-gui"/>
<user username="admin" password="admin" roles="manager-gui,admin-gui"/>
-->
</tomcat-users>
```

```
ubuntu@ip-172-31-40-208: /opt/tomcat
GNU nano 8.7.1 /opt/tomcat/webapps/manager/META-INF/context.xml *
<?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
the License. You may obtain a copy of the License at

    http://www.apache.org/licenses/LICENSE-2.0

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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" ignoreAnnotations="true">
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
    sameSiteCookies="strict" />
  <!--<Valve className="org.apache.catalina.valves.RemoteCIDRValve"
    allow="127.0.0.0/8,::1/128" />-->
  <Manager sessionAttributeValueClassNameFilter="java\\.lang\\.(?:Boolean|Integer|Long|Number|string)|org\\.ap
</Context>
```

```
ubuntu@ip-172-31-40-208: /opt/tomcat
GNU nano 8.7.1 /opt/tomcat/webapps/host-manager/META-INF/context.xml *
<?xml version="1.0" encoding="UTF-8"?>
<!--
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contributor license agreements. See the NOTICE file distributed with
this work for additional information regarding copyright ownership.
The ASF licenses this file to You under the Apache License, Version 2.0
(the "License"); you may not use this file except in compliance with
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distributed under the License is distributed on an "AS IS" BASIS,
WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
See the License for the specific language governing permissions and
limitations under the License.
-->
<Context antiResourceLocking="false" privileged="true" ignoreAnnotations="true">
  <CookieProcessor className="org.apache.tomcat.util.http.Rfc6265CookieProcessor"
    sameSiteCookies="strict" />
  <!--<Valve className="org.apache.catalina.valves.RemoteCIDRValve"
    allow="127.0.0.0/8,::1/128" />-->
  <Manager sessionAttributeValueClassNameFilter="java\\.lang\\. (?:(Boolean|Integer|Long|Number|string)|org\\.ap
</Context>
```

```
ubuntu@ip-172-31-40-208: /opt/tomcat
ubuntu@ip-172-31-40-208: /opt/tomcat$ sudo nano /opt/tomcat/conf/tomcat-users.xml
ubuntu@ip-172-31-40-208: /opt/tomcat$ sudo nano /opt/tomcat/webapps/manager/META-INF/context.xml
ubuntu@ip-172-31-40-208: /opt/tomcat$ sudo nano /opt/tomcat/webapps/host-manager/META-INF/context.xml
ubuntu@ip-172-31-40-208: /opt/tomcat$ sudo systemctl restart tomcat
```

34.238.143.100:8080/manager/html

Iniciar sesión

http://34.238.143.100:8080

Tu conexión con este sitio web no es privada

Nombre de usuario

admin

Contraseña

.....

Cancelar

Iniciar sesión

← → ↻ No es seguro 34.238.143.100:8080/manager/html

 

Gestor de Aplicaciones Web de Tomcat

Mensaje: OK

Gestor

[Listar Aplicaciones](#) [Ayuda HTML de Gestor](#) [Ayuda de Gestor](#) [Estado de Servidor](#)

Ruta	Versión	Nombre a Mostrar	Ejecutándose	Sesiones	Comandos
/	Ninguno especificado	Welcome to Tomcat	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar 2 30 minutos
/docs	Ninguno especificado	Tomcat Documentation	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar 2 30 minutos
/examples	Ninguno especificado	Servlet and JSP Examples	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar 2 30 minutos
/host-manager	Ninguno especificado	Tomcat Host Manager Application	true	0	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar 2 30 minutos
/manager	Ninguno especificado	Tomcat Manager Application	true	1	Arrancar Parar Recargar Replegar Expirar sesiones sin trabajar 2 30 minutos

Desplegar

Desplegar directorio o archivo WAR localizado en servidor

Trayectoria de Contexto (opcional):
 Version (for parallel deployment):
 URL de archivo de Configuración XML:
 URL de WAR o Directorio: [Desplegar](#)

Archivo WAR a desplegar

Seleccione archivo WAR a cargar: [Seleccionar archivo](#) | Ningún archivo seleccionado
[Desplegar](#)

7. Instala la aplicación **sample.war** que se adjunta en el enlace.

← → ↻ No es seguro 34.238.143.100:8080/sample/

 **Sample "Hello, World" Application**

This is the home page for a sample application used to illustrate the source directory organization.

To prove that they work, you can execute either of the following links:

- To a [JSP page](#).
- To a [servlet](#).

8. Instala de forma manual en /var/lib/tomcat11/webapps la aplicación **hello-world.war** que se adjunta en el enlace.

← → ↻ No es seguro 34.238.143.100:8080/hello-world/

Hello World!

It is now Fri May 08 12:01:41 UTC 2026

You are coming from 83.56.0.199

